Project Planning Report: Unsupervised Zero-Day Attack Detection in an Industrial Control System

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***Abstract*—Labelled attack data is scarce in industrial control systems, so a detector must be learned from normal operation alone. This project compares Deep SVDD and a VAE, trained at matched capacity, on their ability to flag attacks they have never seen—including when the plant itself changes. We expect a degradation curve from in-domain to cross-version detection, and a measure of how much of that loss an adaptive decision layer recovers.**

I. Research Question and Expected Outcome

Labelled attack data is scarce in industrial control systems (ICS): attacks are rare and their signatures keep changing, so labelled examples of novel attacks are unavailable at deployment. Hence *one-class* learning: train on normal data only, flag whatever deviates. The project asks one question: **how do Deep SVDD and a VAE, compared at matched capacity on normal ICS telemetry, degrade when the plant itself changes, and can a drift-aware decision layer recover that loss?** Training on HAI 21.03 and testing on 23.05 — a later, differently instrumented plant — makes "zero-day" mean an unseen attack *and* an unseen regime. SVDD compresses normal data into a compact latent hypersphere; a VAE scores by reconstruction probability; to our knowledge the two are rarely compared under a shift of this kind. Expected outcome: a degradation curve from in-domain to cross-version detection, the share of that loss recovered by adaptive thresholding, and a per-attack account of what fails.

II. Datasets

HAI (HIL-based Augmented ICS Security Dataset), CC BY-SA 4.0, open on GitHub (https://github.com/icsdataset/hai) and Kaggle. Training files hold normal operation only; test files hold labelled attacks with documented targets. Primary: HAI 23.05, which ships separate label and attack-summary files alongside the telemetry. Secondary: HAI 21.03, the training domain for the cross-version experiment. The versions differ in tag set (78 vs 86) and label format, so a common-tag subset and two parsers are needed.

III. Available Code and Our Implementation

We reference the official Deep-SVDD implementation[1] (MIT) for the SVDD objective and weight-bounding, and `usad` (KDD'20) / `TranAD` (VLDB'22) as literature comparisons — but will not run them as-is: their image-oriented backbones make their results incomparable with ours.

Our implementation differs in four ways: one shared 1-D CNN encoder so both objectives are compared at matched capacity; one harness applied identically to every model, PCA included; the cross-version protocol these lack; and an extreme-value-theory decision layer replacing a fixed percentile.

IV. Most Relevant Papers

- Ruff et al., *Deep One-Class Classification*, ICML 2018 (PMLR v80).
- Ruff et al., *Rethinking Assumptions in Deep Anomaly Detection*, arXiv:2006.00339, 2020.
- Siffer et al., *Anomaly Detection in Streams with Extreme Value Theory*, KDD 2017.
- Hwang et al., *Time-Series Aware Precision and Recall for Anomaly Detection*, CIKM 2019.
- Kim et al., *Towards a Rigorous Evaluation of Time-series Anomaly Detection*, AAAI 2022.

V. Evaluation

Scores are per sensor, fused per window, so attribution requires no additional computation. Thresholds use streaming extreme value theory (SPOT/DSPOT) instead of a fixed percentile, and all results are reported at a matched **false-alarm budget** in alarms per day. Two threshold arms: frozen from the training domain (pure generalization) and adapting to the test stream (deployment). Metrics: detection rate and mean delay at that budget, PR-AUC, eTaPR, per-attack recall, and whether top-scoring sensors match documented attack targets. Three seeds, mean $\pm$ standard deviation.

VI. Short-Term Plan

Days 1–2: download HAI, read the technical documentation, build loaders and label parsers. Days 3–4: shared encoder with Deep SVDD, then the VAE. Day 5: EVT threshold layer and the false-alarm-budget harness. Day 6: full grid, cross-version runs, per-attack analysis. Day 7: report and poster. Residual-representation and score-fusion variants are optional extensions, time permitting; the main risk — a tag intersection too small to train on — is checked on day 1.

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[1] https://github.com/lukasruff/Deep-SVDD-PyTorch